

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (CE/IT)

May 2014

CE216 Computer Organization & Peripherals

Date: 02.05.2014, Friday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Figures to the right indicate full marks.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) i. Perform addition and subtraction using signed 2's complement system. [02]
(1) $(-29)_{10} + (-49)_{10}$
(2) $(1001)_2 - (101000)_2$
- ii. Differentiate Computer Architecture and Computer Organization. [02]
- (b) A digital computer has a common bus system for 16 registers of 32 bits each. The bus has is constructed with multiplexer. [03]
1. How many selection inputs are there in each multiplexer?
2. What size of multiplexers are needed?
3. How many multiplexer are there in the bus?
- Q - 2 (a) Draw and Explain Common Bus System. Write the step to load value from the memory address is stored in Program Counter register. [06]
- (b) Write an assembly program to circular left shift (four bit position) on given Hexadecimal nos. (1234, 4321) using subroutine call. [04]
- (c) Express the following numbers in IEEE 32-bit floating-point format. [04]
1. $-1/32$
2. 13.1875

OR

- (b) Write an assembly program to multiply two unsigned positive number, each with 16 significant bits, to produce an unsigned double-precision product. [04]
- (c) What is IEEE754 standard? What is infinite range and NaN in IEEE754 standard. [04]
- Q - 3 (a) Divide $(979)_{10}$ by $(31)_{10}$, using 5-bit words. Use division algorithm of unsigned binary integers to find Quotient and Remainder. [06]
- (b) Use the booth algorithm to multiply 30(multiplicand) by -8(multiplier), where each number is represented using 6-bit signed nos. [05]
- (c) What is the difference between a direct and an indirect address instruction? How many references to memory are needed for each type of instruction to bring an operand into a processor register? [03]

OR

- (c) Design a digital circuit that performs the four logic operations of exclusive-OR, exclusive-NOR, NOR and NAND. Use two selection variables. Show the logic diagram of one typical stage. [03]

SECTION - II

- Q - 4 (a) What do you mean by control unit? Which are the two possibility to implement control unit? Explain any of them in brief. [04]
- (b) Convert the following arithmetic expressions from reverse Polish notation to infix notation. [03]

1. $ABCDE+* - /$
2. $ABC*/D-EF/+$
3. $ABCDE*/-+$

- Q - 5 (a) Answer the following: [06]

- (1) Differentiate External interrupts and Internal interrupts.
- (2) Draw and explain Flowchart for first pass of assembler.

- (b) Draw and explain Hardware implementation for signed-magnitude addition and subtraction. [04]
- (c) Design and explain 4-bit combination circuit decremter using four full-adder circuits. [04]

OR

- (b) Draw and explain Interrupt Cycle. [04]

- (c) Explain Flynn's Classification of computing systems. [04]

- Q - 6 (a) Answer the following (Any Four) [12]

- (1) What do you mean by Bus interface? List and explain the functions of Buses.
- (2) An 8-bit register R contain the binary value 10011100 what is the register value after an arithmetic shift right? Starting from the initial number 10011100, determine the register value after an arithmetic shift left, and state whether there is an overflow.
- (3) Explain the important features of RISC based system architecture.
- (4) What do you understand by Fetch cycle, Instruction cycle, Machine cycle, Interrupt acknowledgement.
- (5) Differentiate Register mode and Register indirect mode.

- (b) What is data path and elements of data path. [02]

OR